

Inferring critical transitions from timeseries

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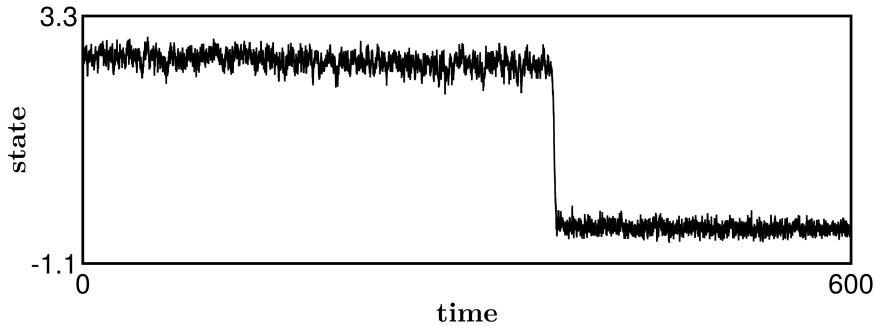
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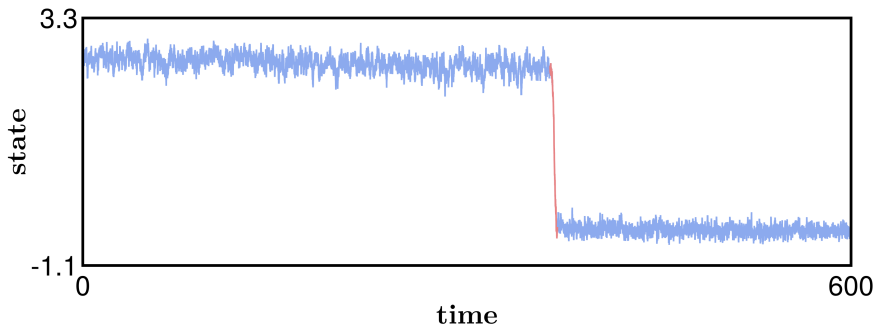
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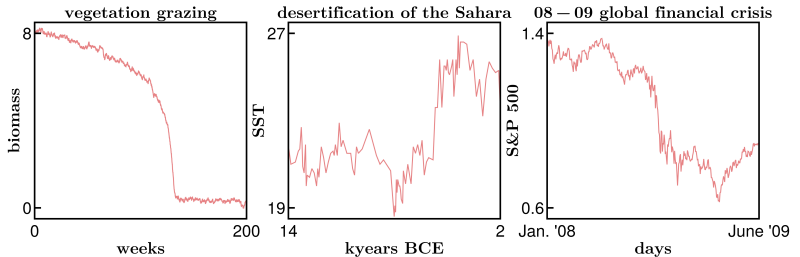
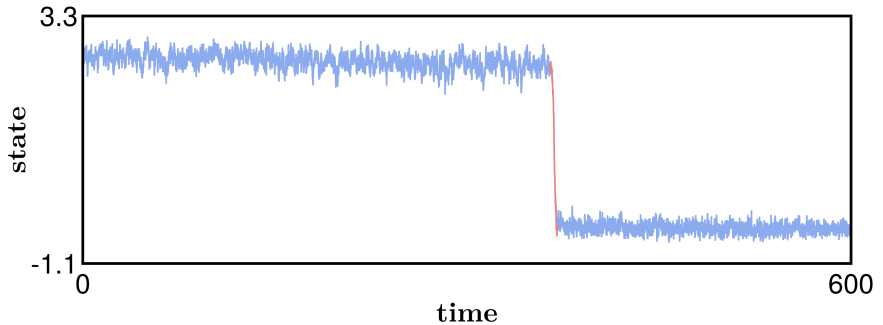
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Dynamical systems that tip

Consider n –dimensional Itô processes with non-autonomous drift and additive noise

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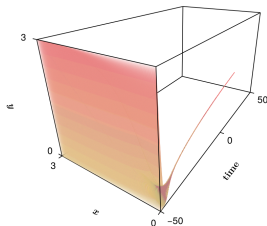
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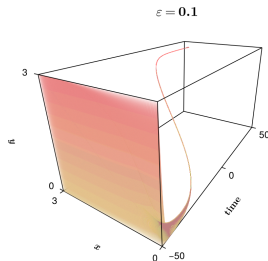
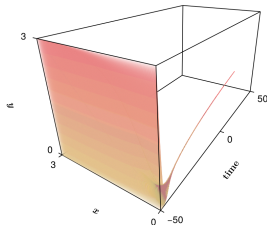
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$$X_t = \theta \int_0^t (\sqrt{-\mu} - X_s) ds + \sqrt{2D} \int_0^t dW_s,$$

which admits a stationary probability density with $\theta = -2\sqrt{-\mu}$ and covariance

$$\text{Cov}[X_s, X_t] = \frac{D}{\theta} e^{-\theta(t-s)}, \quad s < t.$$

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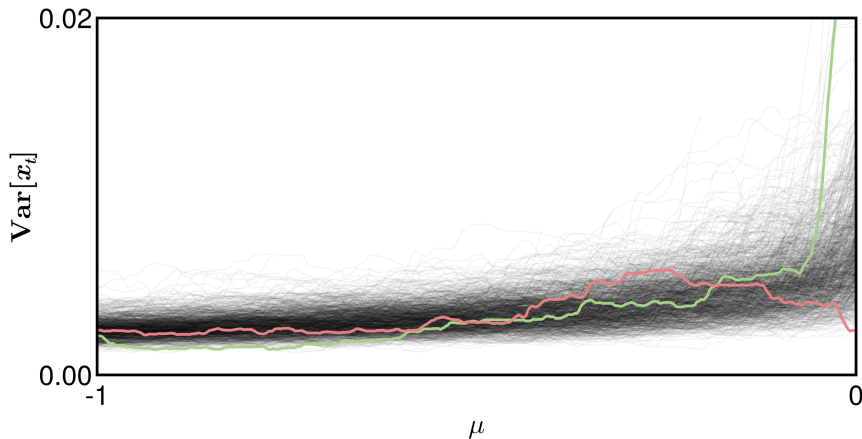
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If we now consider instantaneous time increments (i.e. $t \rightarrow s$) then

$$\text{Cov}[X_s, X_t] \rightarrow \text{Var}[X_t] = \frac{D}{\theta} \rightarrow \begin{cases} 0, & \mu \rightarrow -\infty, \\ \infty, & \mu \rightarrow 0. \end{cases}$$

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$$f(X_t; \mu) = -\mu - X_t^2 \implies \text{Var}[X_t] = \frac{D}{\theta} \rightarrow \begin{cases} 0, & \mu \rightarrow -\infty, \\ \infty, & \mu \rightarrow 0. \end{cases}$$



Entering large deviation theory

Theorem (Freidlin-Wentzell)

The probability of the sample paths $X^D = \{X_t\}_{t=0}^T$ of $dX_t = f(X_t)dt + \sqrt{2D}dW_t$ to be located in a cylinder of radius $\delta > 0$ around any (absolutely continuous) deterministic path ϕ decays as

$$P[\phi] \asymp \exp\left(-\frac{S[\phi]}{2D}\right) \quad \left(\text{or} \quad \lim_{D \rightarrow 0} \log P\left(\sup_{t \in [0, T]} |X_t - \phi(X_0, t)| < \delta\right) = -S[\phi]\right),$$

where the cost of following such path is given by the rate functional

$$S[\phi] = \int_0^T L(\dot{\phi}, \phi, t) dt = \frac{1}{2} \int_0^T \left(\dot{\phi} - f(\phi)\right)^2 dt,$$

In the singular limit $\varepsilon \rightarrow 0$ the most likely path $\phi(X_0, t) \in H^1(\mathbb{R}^d, [0, T])$ is the minimiser of $S[\phi]$ and it satisfies the Euler-Lagrange equations (ELEs)

$$\frac{d}{dt} \left(\frac{\partial}{\partial \dot{\phi}} L(\dot{\phi}, \phi, t) \right) - \frac{\partial}{\partial \phi} L(\dot{\phi}, \phi, t) = 0.$$

Treating tipping points as rare events

Corollary

Consider a 1-dimensional, conservative vector field f (i.e. $\exists V$ s.t. $f = -\nabla V$) with a unique, stable fixed point x_0 . Then the probability for hitting a target x scales as $\exp(V(x_0) - V(x))$.

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Consider a vector field $f(x; \mu)$ with equilibria $x_{\text{stb}, \text{uns}}$.

If the vector field is conservative then its potential $V(x; \mu)$ has a local minima at $a \equiv x_{\text{stb}}$ and a local maxima at $b \equiv x_{\text{uns}}$.

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Approaching a saddle-node bifurcation ($\mu \rightarrow \mu_{\text{bif}}$) we have $|x_{\text{stb}} - x_{\text{uns}}| = |a - b| \rightarrow 0$, meaning that

$$V(b) - V(a) =: \Delta V \rightarrow 0 \implies \exp(-\Delta V) \rightarrow 1.$$

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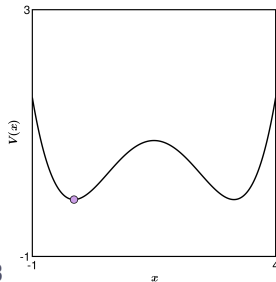
We can use the LDP as an EWS!

$$\exp(-\Delta V) = \begin{cases} 0, & \text{if } |\mu - \mu_{\text{bif}}| \rightarrow \infty, \\ 1, & \text{if } |\mu - \mu_{\text{bif}}| \rightarrow 0. \end{cases}$$

Transition probabilities

Given a 1-dimensional conservative system $dX_t = -V'(X_t; \mu)dt + \sqrt{2D}dW_t$ the transition probability $p(X_t, t|X_0, 0)$ satisfies the Fokker-Plank equation

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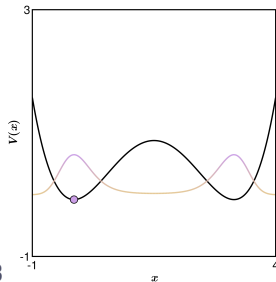
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In the singular limit $\varepsilon \rightarrow 0$ the transition probability solves the second-order IVP

$$\begin{cases} Dp'' + V''p + V'p' = 0, \\ p'(\pm\infty) = 0. \end{cases}$$

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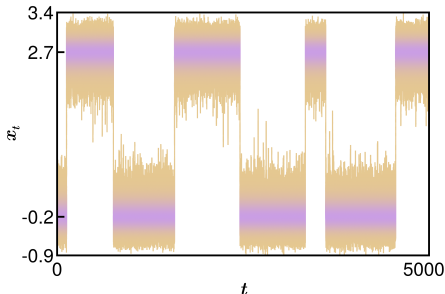
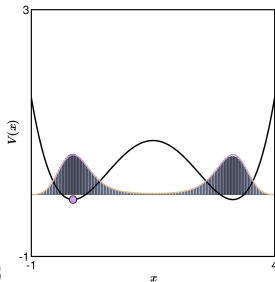
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$$\hat{p}_h(x) = \sum_{k=1}^M \hat{p}_k \mathbf{1}_{B_k} = \frac{1}{Nh} \sum_{k=1}^M \mathbf{1}_{B_k}(x) \sum_{n=1}^N \mathbf{1}_{B_k}(X_n),$$

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Proposition

The fluctuations between the unbiased density estimator $\hat{p}_h$ and the bin-averaged histogram $p_h = \sum_{k=1}^M p_k \mathbf{1}_{B_k}$, $p_k = \frac{1}{h} \int_{B_k} p(x)dx$ are asymptotically Gaussian

$$\sqrt{N}(\hat{p}_k - p_k) \xrightarrow{d} \mathcal{N}\left(0, \frac{\pi_k(1 - \pi_k)}{h^2}\right), \quad k = 1, \dots, M,$$

where $\pi_k := Pr[x \in B_k] = hp_k$ defines the bin probability.

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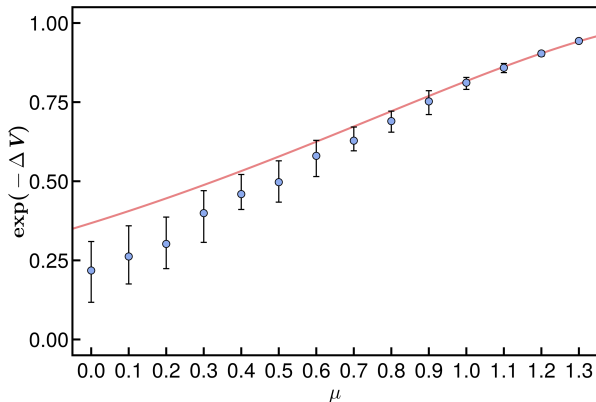
Given the (empirical) histogram $\{\hat{x}_k, \hat{p}_k\}$ over the bins' center points $\hat{x}_k$, find

$$\boldsymbol{\theta}^* = (c_j)_{j=0,\dots,3} = \operatorname{argmin}_{\boldsymbol{\theta} \in \mathbb{R}^4} \sqrt{\sum_{k=1}^M (\exp(-U(\hat{x}_k; \boldsymbol{\theta})) - \hat{p}_k)^2}.$$

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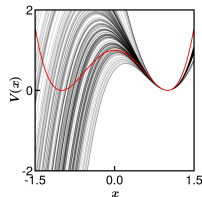
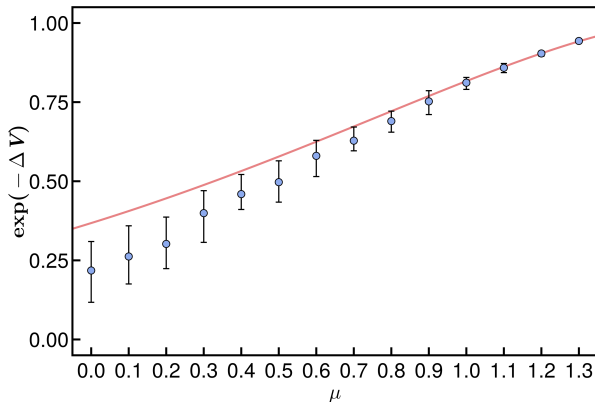
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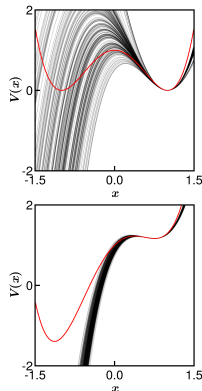
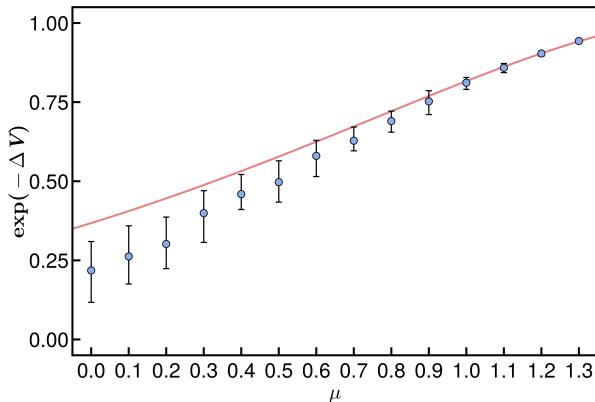
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Thank you!

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